

Content Management in Health Care

Automating Billing Policy Updates with AI

Cotiviti Intern Assessment | Topic 3

1. The Problem

Every January 1st, the rules change. The AMA released **420 updates to the CPT code set for 2025 alone**, 270 new codes, 112 deletions, and 38 revisions (AMA, 2024). CMS also updates its coverage determinations and telehealth guidance on a rolling basis throughout the year (CMS MLN901705, 2025). For Cotiviti, each update requires a human analyst to read the new policy PDF, compare it to the prior version, identify what changed, and rewrite the internal rules that govern how claims get evaluated. Across thousands of policies, **this process consumes weeks of analyst time annually**, time spent reading documents rather than making decisions. Hospital administrative costs hit \$687 billion in 2023, and billing errors driven by outdated rules contribute directly to the \$18 billion hospitals spent overturning claim denials in 2025 alone (Trilliant Health, 2024; AHA, 2025).

2. Relevant Trends

2.1 LLMs can now extract rules from policy language reliably

Retrieval-augmented generation (RAG), has shown strong results in medical guideline interpretation. **One study found accuracy improved from 43 percent to 99 percent when RAG was applied to hepatitis C treatment guidelines** (Kresevic et al., 2024, as cited in PMC, 2025). The key is grounding: the model cites the exact source sentence, which makes its output verifiable.

2.2 Policy volume is growing and the cost of errors is rising

CPT updates now run to hundreds of changes per year, and CMS telehealth policy has shifted significantly each year since the COVID-19 public health emergency ended in 2023. **Hospitals spent \$43 billion in 2025 collecting payments insurers already owed**, a problem that starts upstream with rules being misapplied (AHA, 2025). Faster policy cycles mean more chances for a rule to be missed or misread.

3. Opportunities and Threats

3.1 Opportunities

An AI pipeline that reads a new policy PDF, compares it to the prior version, and produces a structured change summary compresses the analyst's update workflow from days to hours. Every extracted rule links back to its source sentence, which makes the output **auditable and defensible during CMS and OIG reviews** - something that manually written rules often are not.

3.2 Threats

The primary threat is hallucination: an LLM extracting a rule that does not accurately reflect the source document. In healthcare billing, **a faulty rule means wrongful denials or inappropriate approvals with financial and legal consequences**. Policy language is also

frequently ambiguous, and LLMs tend to resolve ambiguity silently. A production system must flag low-confidence extractions for human review rather than proceeding automatically.

4. Technology: Proof of Concept and Production Path

4.1 What was built

A three-agent pipeline was built in Python using pdfplumber for PDF parsing and DeepSeek via the OpenRouter API. **Agent 1 extracts billing rules from a policy PDF as structured JSON with source citations. Agent 2 compares two rule sets and produces a plain-English diff report. Agent 3 evaluates test claims against the current rules and explains each decision.** LangChain and vector databases were excluded intentionally, for two documents, they add complexity without adding capability for demo.

4.2 What production would look like

At Cotiviti's scale, x thousands of LCDs, NCDs, and payer contracts - **LlamaIndex would replace the direct extraction layer.** It builds a persistent semantic index across all policy documents so the LLM retrieves only the relevant chunks for a given query instead of reprocessing full documents each time. The citation layer stays identical: every rule traces back to the exact sentence that justifies it, so a reviewer can verify without reading the full document.

5. Conclusion and Recommendation

The bottleneck in healthcare content management is not that billing policies are complex. It is that **converting policy updates into working rules is still a manual reading exercise.** Current LLM technology is capable of automating the reading, extraction, and comparison steps while keeping a human as the final approver.

Cotiviti should invest in an internal policy ingestion pipeline starting with remote patient monitoring. RPM policy has changed substantially year over year, CMS publishes it in consistently formatted documents that parse cleanly, and RPM claim volume is growing as telehealth expands. **The return on getting RPM rules right increases every quarter.** The goal is not to replace the analyst. It is to change what they spend their time on: from reading PDFs to reviewing AI-generated summaries and making the final call.

References

- American Hospital Association. (2025). Costs of caring. <https://www.aha.org/costsofcaring>
- American Hospital Association. (2024, September 10). Skyrocketing hospital administrative costs, burdensome commercial insurer policies impacting patient care. <https://www.aha.org/guidesreports/2024-09-10-skyrocketing-hospital-administrative-costs>
- American Medical Association. (2024, September). AMA releases CPT 2025 code set. <https://www.ama-assn.org/press-center/ama-press-releases/ama-releases-cpt-2025-code-set>
- American Medical Association. (2025, September). AMA releases CPT 2026 code set. <https://www.ama-assn.org/press-center/ama-press-releases/ama-releases-cpt-2026-code-set>
- Centers for Medicare and Medicaid Services. (2025, December). Telehealth and remote monitoring (MLN901705). <https://www.cms.gov/files/document/mln901705-telehealth-remote-monitoring.pdf>
- Centers for Medicare and Medicaid Services. (2024, November). Annual update to the list of CPT/HCPCS codes effective January 1, 2025. <https://www.cms.gov/medicare/regulations-guidance/physician-self-referral/list-cpt-hcpcs-codes>
- Manz, C. R., et al. (2025). Retrieval augmented generation for large language models in healthcare: A systematic review. PMC. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12157099/>
- Trilliant Health. (2024, March). Hospital administrative expenditures exceed direct patient care by nearly 2x. <https://www.trillianthealth.com/market-research/studies/hospital-administrative-expenditures-exceed-direct-patient-care-by-nearly-2x>
- University of Texas Health Science Center. (2025). 2025 coding compliance changes. <https://med.uth.edu/mshbc/coding-compliance-overview/2025-coding-compliance-changes/>
- Xie, J., et al. (2025). Retrieval augmented generation evaluation for health documents. arXiv. <https://arxiv.org/pdf/2505.04680>